

Download Checklist — MYRRHA QPLANT Cryogenic Leak Rate Dashboard v4.0.0

Use this checklist to ensure you have everything needed for a complete project handover.

Essential Files (Always Needed)

These files are required to understand, run, and extend the project.

Version & Metadata

- [] `VERSION` — Plain text version (4.0.0) — < 1 KB
- [] `VERSION.json` — Structured version metadata — < 1 KB
- [] `CHANGELOG.md` — Full version history — ~5 KB
- [] `README.md` — Project overview and quick start — ~2 KB

Handover Package

- [] `HANDOVER.md` — Master handover document — ~8 KB
- [] `ARTIFACTS.yaml` — Machine-readable artifact catalog — ~10 KB
- [] `DOWNLOAD_CHECKLIST.md` — This file — ~4 KB
- [] `REBUILD_GUIDE.md` — Rebuild instructions — ~5 KB

Configuration (SSoT)

- [] `data/config.yaml` — Single Source of Truth — ~6 KB
- [] `data/config.json` — Machine-readable SSoT — ~3 KB
- [] `data/config.md` — Documentation tables — ~3 KB
- [] `data/compressor_specs.json` — Compressor data — ~4 KB
- [] `data/helium_properties.json` — He properties — ~10 KB
- [] `data/leak_classes.json` — Leak classifications — ~1 KB
- [] `data/scenarios.json` — System scenarios — ~2 KB
- [] `data/valve_candidates.json` — Valve data — ~1 KB
- [] `data/source_anchors.json` — Source references — ~1 KB
- [] `data/standards_compliance.json` — Standards framework — ~34 KB
- [] `data/material_costs.json` — Cost data — ~2 KB
- [] `data/operating_conditions.json` — Operating conditions — ~1 KB
- [] `data/meca_inox_specs.json` — Valve specs — < 1 KB
- [] `data/swagelok_specs.json` — Valve specs — < 1 KB

Source Code

- [] `src/calc_leak_rate.py` — Core physics engine — ~8 KB
- [] `src/config_loader.py` — SSoT config loader — ~4 KB
- [] `src/generate_dashboard.py` — Dashboard generator — ~20 KB
- [] `src/build_all.py` — Build orchestrator — ~5 KB
- [] `src/build_dashboard.py` — Landing page generator — ~5 KB

- [] `src/build_handover.py` — Handover doc generator — ~6 KB
- [] `src/manifest.py` — Hashing & idempotent writes — ~3 KB
- [] `src/compressor_reliability.py` — Compressor models — ~8 KB
- [] `src/liquid_he_loss.py` — Liquid He loss calc — ~4 KB
- [] `src/wcs_scenarios.py` — WCS scenarios — ~5 KB
- [] `src/generate_visuals_v3.py` — Chart generator — ~15 KB
- [] `src/generate_standards_stats.py` — Standards generator — ~12 KB
- [] `src/monte_carlo.py` — Monte Carlo engine — ~6 KB
- [] `src/risk_model.py` — Risk model — ~4 KB
- [] `src/materials_db.py` — Materials database — ~5 KB
- [] `src/verify_triage.py` — Triage compliance — ~3 KB
- [] `src/__init__.py` — Package marker — < 1 KB
- [] `src/calculations/__init__.py` — Package marker — < 1 KB
- [] `src/plotting/__init__.py` — Package marker — < 1 KB

Tests

- [] `tests/test_engineering.py` — ~3 KB
- [] `tests/test_calculations.py` — ~2 KB
- [] `tests/test_config_loader.py` — ~2 KB
- [] `tests/test_data_integrity.py` — ~2 KB
- [] `tests/test_build_outputs.py` — ~2 KB
- [] `tests/test_outputs.py` — ~2 KB
- [] `tests/conftest.py` — ~1 KB

Build Scripts

- [] `setup.sh` — Environment setup — ~2 KB
- [] `build.sh` — Full build — ~2 KB
- [] `validate.sh` — Test & verify — ~2 KB
- [] `package.sh` — Create distribution — ~2 KB
- [] `Makefile` — Make targets — ~1 KB
- [] `requirements.txt` — Python dependencies — < 1 KB

Web Assets

- [] `assets/style.css` — Dashboard styling — ~3 KB
- [] `assets/triage.js` — Interactive controls — < 1 KB

CI/CD

- [] `.github/workflows/build.yml` — CI pipeline — ~1 KB
- [] `.gitignore` — Git ignore rules — < 1 KB

Subtotal (essential): ~108 KB source + ~62 KB data = **~170 KB**

Generated Files (Can Be Rebuilt)

These are generated by `./build.sh` and `./validate.sh`. You can skip them if you plan to rebuild.

Documentation (HTML)

- [] docs/index.html — Landing page — ~5 KB
- [] docs/index_v4_0.html — 40-slide navigator — ~80 KB
- [] docs/STAKEHOLDER_PRESENTATION.html — Stakeholder deck — ~20 KB
- [] docs/dashboard.html — Interactive dashboard — ~4 KB
- [] docs/calculations.html — Calculation pages — ~27 KB
- [] docs/executive_summary.html — Executive summary — ~3 KB
- [] docs/rtm_traceability.html — RTM page — ~8 KB
- [] docs/handover.html — Handover (HTML) — ~9 KB
- [] docs/handover.pdf — Handover (PDF) — ~26 KB
- [] docs/triage_compliance.html — Compliance report — ~92 KB
- [] docs/compressors/*.html — 2 pages — ~16 KB
- [] docs/liquid_he/*.html — 1 page — ~7 KB
- [] docs/standards/*.md + *.csv + *.json — 4 files — ~10 KB

Visualizations

- [] docs/plots/*.html — 5 core Plotly charts — ~47 KB
- [] docs/visualizations/*.html — 18 v3.0 charts — ~2.2 MB
- [] docs/visualizations/data/*.csv — 18 backing data files — ~525 KB
- [] docs/visualizations_v3/*.html — 22 v3.1 charts — ~1.5 MB

Data Exports

- [] docs/data_exports/*.csv + *.json — Full conversion grids — ~70 KB
- [] docs/tables/*.html — HTML tables — ~6 KB
- [] docs/tables_v3/*.csv — Extended CSV tables — ~1.5 MB
- [] docs/statistical/*.json — Statistical results — ~2 KB

Build State

- [] docs/manifest.json — Build manifest — ~15 KB
- [] docs/backlog.json — Backlog items — ~3 KB
- [] OUTPUT_MANIFEST.json — Output manifest — ~5 KB

Test Reports

- [] dist/test-report.html — HTML test report — ~13 KB
- [] dist/test-results.json — Test results JSON — < 1 KB
- [] dist/junit.xml — JUnit XML — ~2 KB

Distribution

- [] dist/handover.zip — Complete bundle — ~2.7 MB
- [] dist/handover.zip.sha256 — Integrity checksum — < 1 KB

Subtotal (generated): ~6.5 MB

Recommended Zip Contents

For a **minimal handover** (rebuilds everything from source):

```

cryo_leak_rate_dashboard_source.zip
├── data/           (~108 KB)
├── src/            (~120 KB)
├── tests/         (~15 KB)
├── assets/        (~4 KB)
├── *.sh, Makefile (~10 KB)
├── VERSION, VERSION.json, CHANGELOG.md
├── HANDOVER.md, ARTIFACTS.yaml, DOWNLOAD_CHECKLIST.md, REBUILD_GUIDE.md
├── README.md, requirements.txt, .gitignore
├── .github/workflows/build.yml
└──
Total: ~300 KB

```

For a **complete handover** (includes all generated outputs):

```

cryo_leak_rate_dashboard_full.zip
├── (everything in minimal)
├── docs/           (~6.9 MB)
├── outputs/        (~varies)
├── dist/           (~2.8 MB)
├── tables/         (~20 KB)
├── traceability/   (~5 KB)
├── source/         (~10 KB)
└──
Total: ~10 MB

```

Pre-built bundle available: `dist/handover.zip` (~2.7 MB)

Verification After Download

```

# 1. Verify zip integrity (if using dist/handover.zip)
sha256sum -c dist/handover.zip.sha256

# 2. Setup environment
./setup.sh

# 3. Run all tests
./validate.sh

# 4. Expected result: "22 passed"

# 5. Check build state
python3 -c "import json; m=json.load(open('docs/manifest.json')); print(f'Version: {open(\"VERSION\").read().strip()}, Status: {m[\"build\"][\"status\"]}, Tests: {m[\"tests\"][\"summary\"]}')"

```